Data Analytics Capstone

Multi-Modal MRI-Based Glioma Tumor Classification Using Machine Learning and Deep Learning

Final Synopsis

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Introduction

Today as we can interpret based on many medical reports, gliomas are the most common and aggressive type of primary brain tumor, which can be seen sometimes on news channels also, significantly compromising patient survival and quality of life. Here an early and accurate diagnosis is essential for treatment planning, prognosis estimation, and improving clinical outcomes (Zhao et al., 2022). Conventionally, diagnosis of glioma is made based on radiological imaging/MRI and histological processing, which are usually labor-intensive, subjective, and in most cases invasive (Jekel et al., 2022). Such limitations demonstrate the necessity of automated, efficient, and non-invasive diagnostic channels.

In this framework, Magnetic Resonance Imaging (MRI) provides the most comprehensive structural and functional information for normal and neoplastic tissue which renders it essential in neuro-oncology. However, manual interpretation of MRI scans does rely on expert radiologists and has interobserver variability (Merkaj et al., 2022). Automated tumor diagnosis and classification using machine learning (ML) and deep learning (DL) are all popular with sophisticated data analytics.

Recent work suggests that monocentric MRI fails to fully approximate the complexity of tumor phenotypes. To obtain complementary information about tumor structure, edema, and tissue composition, multi-modal MRI, including T1-weighted (T1), contrast-enhanced T1 (T1CE: post-contrast T1), T2-weighted (T2), and FLAIR sequences, is used. We can obtain more complete tumor characteristics by integrating these modalities.

The objective of the study is to create an effective tumor classification system using multi-modal MRI data in combination with comparative performance analysis of machine learning and deep learning models. The results are expected to facilitate the implementation of smart healthcare diagnostic systems.

Scope and Objectives

Problem Statement

Due to the inter-tumor heterogeneities, varying imaging conditions, and complexity of brain structure, precisely classifying glioma tumors directly from MRI data is still a challenging task. Current methods either depend on few imaging modalities or do not sufficiently combine the structured and unstructured data. In this research, we resolve these limitations by developing a comprehensive classification framework on multi-modal MRI data and exploring both machine learning as well as deep learning approaches to implementing the same.

Scope of the Study

This study is based on the following:

1. We performed analysis on multi-modal MRI data (T1, T1CE, T2 and FLAIR)
2. Segmentation masks for tumor regions facilitating feature extraction
3. Machine learning modeling with structured features
4. Application of deep learning models to MRI images
5. Comparative analysis of ML vs. DL approaches

It is a binary classification study (tumor vs non-tumor), not clinical deployment.

Feasibility

Feasibility of the project within timelines as below:

1. Availability of dataset

2. Implementation in Python and Google Colab

3. Pre-trained deep learning model usage

4. Moderate computational requirements

Ethical Considerations are essential

1. Dataset is anonymized

2. This Example Does not Use any Personal Patient Data

3. Research complies with ethical standards

Objectives

1. For classification of tumors, we are creating machine learning models

2. Deep learning models to analyze images from an MRI

3. Assessing the impact of multi-modal MRI integration

4. For evaluating the performance of ML vs. DL models

Research Questions and Hypotheses are as follows:

RQ1: Given multi-modal MRI features, how well can machine learning models classify glioma tumors?

H1: High classification accuracy for machine learning models.

RQ2: Now, how well do deep learning models work on MRI image data?

H2: Deep learning models will have intermediate accuracy.

RQ3: Does integration of multi-modal MRIs improve classification accuracy?

H3: Multi-modal MRI achieves superior classification performance.

RQ4: On The Contrast, Which Is Better: Machine Learning or Deep Learning?

H4: Machine Learning Models are Better than Deep Learning Models for Properly Structured Feature Engineering.

Sample Size Calculation

The sample size is calculated using the confidence interval formula:

$$N=\frac{Z^2.p(1-p)}{e^2}$$

Where:

Z = 1.96

p = 0.5

Value of e = 0.05

N=384

RQ	Method	Parameters	Minimum Sample
RQ1	CI	Z=1.96, e=0.05	384
RQ2	CI	Z=1.96, e=0.05	384
RQ3	Power	α=0.05, Power=0.8	400

RQ4	Comparative	Balanced data	400

The dataset used here (containing~2000 samples) exceeds this requirement, ensuring statistical reliability. Its really a big dataset.

Data Description

Dataset Overview

1. Dataset her contains a big UTSW Glioma Dataset (https://www.cancerimagingarchive.net/collection/utsw-glioma/)

2. Patients: ~625

3. MRI modalities are: T1, T1CE, T2,and FLAIR

Data Processing

1. MRI images converted from NIfTI format

2. Tumor segmentation applied

3. Feature extraction performed

4. Data normalized and cleaned

Data Dictionary

Feature	**Description**
t1_mean	Mean intensity
t1_std	Standard deviation
t2_max	Maximum intensity

flair_range	Intensity range
label	Tumor / Non-tumor

GitHub: https://github.com/kavidha123/great-learning

Analytic Approach

Overview

The analytic technique used in this study is a hybrid combination of machine learning (ML) and deep learning (DL) methods to assess glioma tumor classification based on multi-modal MRI data. It is based on the combination of a feature-based model and image-based learning, offering a full view on classification performance. For each research question, a corresponding analytical strategy is aligned to achieve the study aims.

RQ1: Machine Learning Performance

Research Question:

To what level can multimodal MRI features classify glioma tumors with machine learning models?

Solution Approach:

Multi-modal MRI data contain three fixed structure features (T1, T1CE, T2, and FLAIR), which were respectively demarcated with segmentation masks —mean, standard deviation, maximum, minimum, and range of intensity values from tumor regions. The models used: Logistic Regression, SVMs, K-Nearest Neighbor (KNN), Random Forest, Gradient Boosting, and XGBoost.

Justification:

For structured datasets, machine learning models are potent even when features are created on specific domains. Here ensemble methods like Random Forest and XGBoost capture more complex and non-linear relationships.

Expected Outcome:

The use of informative features gives high classification accuracy.

RQ2: Deep Learning Performance

Research Question:

How well do deep learning models perform on MRI image data?

Solution Approach:

Based on the research, we can see that convolutional neural networks (CNNs) here use two-dimensional slices converted from MRI scans for input from the database. Here we use models that include ResNet18, EfficientNet-B0, and MobileNet. Transfer learning: Use pre-trained ImageNet weights.

Justification:

Here deep learning models automatically learn various hierarchical features from raw images without manual feature extraction. For image classification tasks, architectures like ResNet and EfficientNet are suitable.

Expected Outcome:

Due to the limited size of datasets, moderate performance is anticipated.

RQ3: Multi-Modal MRI Impact

Research Question:

Does classification generalization improve with multi-modal MRI integration?

Solution Approach:

Features are extracted for each modality (T1, T1CE, T2, FLAIR) separately and merged into a single feature vector using feature fusion.

Justification:

Each modality reflects distinct features of the tumor. Having a mix creates a better representation and leads to better model performance.

Expected Outcome:

The integration of multiple modalities can lead to a massive boost in classification accuracies.

RQ4: ML vs DL Comparison

Research Question:

How do machine learning and deep learning approaches perform relatively?

Solution Approach:

Using a comparative framework, ML models are built on structured features, while DL uses the raw MRI images. They are both assessed by similar metrics.

Justification:

Based on the research process,we can see that ML models harness feature engineering, while DL models use automatic feature learning. When comparing the two, you can better understand the strengths and limitations of each.

Expected Outcome:

Structured data can be a benefit facing machine learning models vs. deep learning ones (which will likely perform better on this dataset anyway, assuming the dataset is large enough).

Evaluation Metrics (APPLIES TO ALL RQs)

The performance of all models will be evaluated using the following metrics:

$$\text{Accuracy}=\frac{\text{TP+TN}}{\text{TP+TN+FP+FN}}$$

$$\text{Precision}=\frac{\text{TP}}{\text{TP+FP}}$$

$$\text{Recall}=\frac{\text{TP}}{\text{TP+FN}}$$

$$\text{F1}=2.\frac{\text{Precision}\cdot\text{Recall}}{\text{Precision+Recall}}$$

ROC-AUC

Measures the ability of the model to distinguish between classes across thresholds.

Expected Outcomes

1. ML models expected to perform better

2. DL models expected to perform moderately

3. Multi-modal MRI expected to improve accuracy

4. Ensemble models expected to perform best

Recommendations and Applications

Target Users

1. Radiologists

2. Hospitals

3. AI healthcare systems

Applications

1. Automated tumor detection

2. Clinical decision support

3. Early diagnosis systems

Literature Review

Glioma as we know with complicated biological behavior is the most common primary brain tumor, but it has high invasiveness and drug resistance. Recent publications have underscored the role of machine learning (ML) in glioma classification, grading, and prognostication. Studies have reported discriminative accuracy of ML models, such as support vector machines and deep-learning approaches, that is near the highest possible when differentiating glioma from brain metastasis and in grading tumor aggressiveness; however, small datasets rarely followed by external validation and poor reporting standards limit their applicability to the clinic (Jekel et al., 2022; Merkaj et al., 2022). The above developments have further permitted non-invasive tumor grading due to quantification of imaging features in various domains with radiomics-based approaches, although their data size and heterogeneity are limited (Kumar et al., 2023).

It is also possible to make survival predictions directly from ML models trained on large-scale clinical datasets, with performance suggesting that this may be a valid alternative to more traditional approaches utilising clinical variables as prognostic factors (Zhao et al., 2022). At the same time we see early efforts with new drug therapy, example ONC201 provides significant clinical response in recurrent glioma, while combination strategies such as oncolytic virus plus chemotherapy improved tumor control; suggesting progress still being made in the armamentarium of therapy (Arrillaga-Romany et al., 2024; Vasileva et al., 2024).

Biologically, there are solid mechanistic links between tumor invasion and the accompanying tumor microenvironment (including hypoxia and extracellular matrix interactions, as well as the involvement of immune cells) with glioma progression and therapy resistance (Ohnishi, 2024; Tamai et al., 2022). Moreover, neuron–glioma interactions emphasize the reconstruction of neural circuits by tumor cells as well as neurotransmitter signaling, which drives tumor expansion (Hua et al., 2022). GSCs are important for the tumor initiation, recurrence, and treatment resistance of glioma and are driven by key molecular pathways (Notch, PI3K/AKT, and STAT3) (Agosti et al., 2024).

In conclusion we can say, the integration of ML-based tools for diagnostic determination with biological insights and guided therapies may serve as a future strategy towards precision medicine in glioma management.

References

Agosti, E., Antonietti, S., Ius, T., Fontanella, M. M., Zeppieri, M., & Panciani, P. P. (2024). Glioma stem cells as promoter of glioma progression: A systematic review of molecular pathways and targeted therapies. *International Journal of Molecular Sciences, 25*(14), 7979. https://doi.org/10.3390/ijms25147979

Arrillaga-Romany, I., Gardner, S. L., Odia, Y., Aguilera, D., Allen, J. E., Batchelor, T., Butowski, N., … & Wen, P. Y. (2024). ONC201 (dordaviprone) in recurrent H3 K27M–mutant diffuse midline glioma. *Journal of Clinical Oncology, 42*(13), 1542–1552. https://doi.org/10.1200/JCO.23.01134

Hua, T., Shi, H., Zhu, M., Chen, C., Su, Y., Wen, S., Zhang, X., Chen, J., Huang, Q., & Wang, H. (2022). Glioma-neuronal interactions in tumor progression: Mechanism, therapeutic strategies and perspectives (Review). *International Journal of Oncology*. https://doi.org/10.3892/ijo.2022.5394

Jekel, L., Brim, W. R., von Reppert, M., Staib, L., Cassinelli Petersen, G., Merkaj, S., Subramanian, H., Zeevi, T., Payabvash, S., & Aboian, M. S. (2022). Machine learning applications for differentiation of glioma from brain metastasis—A systematic review. *Cancers, 14*(6), 1369. https://doi.org/10.3390/cancers14061369

Kumar, A., Jha, A. K., Agarwal, J. P., Yadav, M., Badhe, S., Sahay, A., Epari, S., Sahu, A., Bhattacharya, K., & Goda, J. S. (2023).

Machine-learning-based radiomics for classifying glioma grade from magnetic resonance images of the brain. *Journal of Personalized Medicine, 13*(6), 920.

https://doi.org/10.3390/jpm13060920

Merkaj, S., Bahar, R. C., Zeevi, T., Lin, M., Ikuta, I., Bousabarah, K., Cassinelli Petersen, G. I., Staib, L., Payabvash, S., & Aboian, M. S. (2022).

Machine learning tools for image-based glioma grading and the quality of their reporting: Challenges and opportunities. *Cancers, 14*(11), 2623.

https://doi.org/10.3390/cancers14112623

Ohnishi, T. (2024).

Current status and future perspective in glioma invasion research. *Brain Sciences, 14*(4), 309.

https://doi.org/10.3390/brainsci14040309

Tamai, S., Ichinose, T., Tsutsui, T., Tanaka, S., Garaeva, F., Sabit, H., & Nakada, M. (2022).

Tumor microenvironment in glioma invasion. *Brain Sciences, 12*(4), 505.

https://doi.org/10.3390/brainsci12040505

Vasileva, N., Ageenko, A., Byvakina, A., Sen'kova, A., Kochneva, G., Mishinov, S., Richter, V., & Kuligina, E. (2024).

The recombinant oncolytic virus VV-GMCSF-Lact and chemotherapy drugs against human glioma. *International Journal of Molecular Sciences, 25*(8), 4244. https://doi.org/10.3390/ijms25084244

Zhao, R., Zhuge, Y., & Krauze, A. V. (2022).

Machine learning based survival prediction in glioma using large-scale registry data. *Health Informatics Journal*. https://doi.org/10.1177/14604582221135427